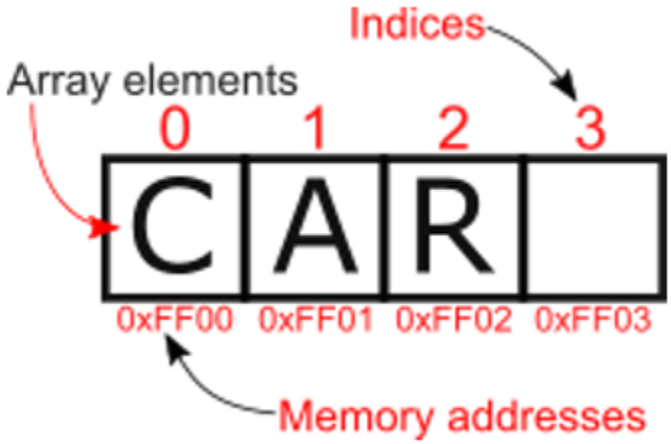


Figure 1.1 The (simplified) internal representation of an array.
Each element of the array in the picture corresponds to a byte of memory[3], whose address is shown below it. On top of each element, instead, its index is shown.
An array is stored as a contiguous block of memory, and each element's address can be obtained by adding its index within the array to the offset of the first element.

For instance, the 4th character of the array (`array[3]`, empty in the figure), has address `0xFF00 + 3 = 0xFF03`.

Data structures can be abstract, or concrete.

- An abstract data type (ADT) specify the operations that can be performed on some data and the computational complexity of those operations. No details are provided on how data are stored or how physical memory is used.